Studying News Coverage through Embeddings Networks: Opportunities and Challenges for Historical Research

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Abstract

Does network analysis and visualization enable effective exploration of a large historical text corpora? And more specifically, does it allow us to understand the overall picture of news coverage in press archives more effectively than other methods of statistical and visual exploration? Press archive collections are indeed large bodies of text that are relatively uniform (and therefore comparable with each other and over time) and are increasingly being digitized (Bunout et al., 2023). This makes it an ideal testing ground for exploring the potential of historical network analysis: following on from our reflections on the study of networks of named entities in the press (Grandjean, 2025a), this contribution therefore has primarily an epistemological ambition. Even if they are not part of the canonical list of standard historical network models (Grandjean, 2025b), are semantic similarity networks based on embeddings any good for historians?
A recent machine learning Natural Language Processing (NLP) tool, which complements the topic models already used for press corpus analysis (Murugaraj et al., 2025), text embedding models "promise to map the meaning or content of sentences and documents to useful numerical vector representations" (Opitz et al., 2025) that enable the distance between two texts to be calculated. These distances make it possible to produce UMAP visualizations that flatten semantic complexity to reveal text clusters, but it is also primarily a network of similarities between texts, which can be analyzed as a k-nearest neighbors' network (Jacomy, 2025), as in Fig. 1 (left). UMAP visualizations of text embeddings are notoriously difficult to interpret, although they can be useful when the texts are highly standardized, such as in corpora of news articles (Michelet and Grandjean, 2026). However, it appears that enriching these visualizations with links to the nearest neighbors (and thus transforming the scatter plots into network visualizations) greatly helps in understanding how the dimensionality reduction algorithm works and in interpreting the results visually and interactively.

But can these similarities be explained/interpreted (Optiz et al., 2025)? And more importantly, can these networks be "translated" into the language of the archival texts from which they are derived (Grandjean and Jacomy, 2019)? In the case of press articles, what type of sampling should be carried out so that the clusters obtained tell us something about the media coverage of an historical phenomenon? To what extent can historians intervene with research questions that are not solely data-driven, for example by integrating another archival source to see where it fits into the structure obtained (Fig. 1 right) ? And finally, are the results robust, or is the tool primarily an exploration interface?

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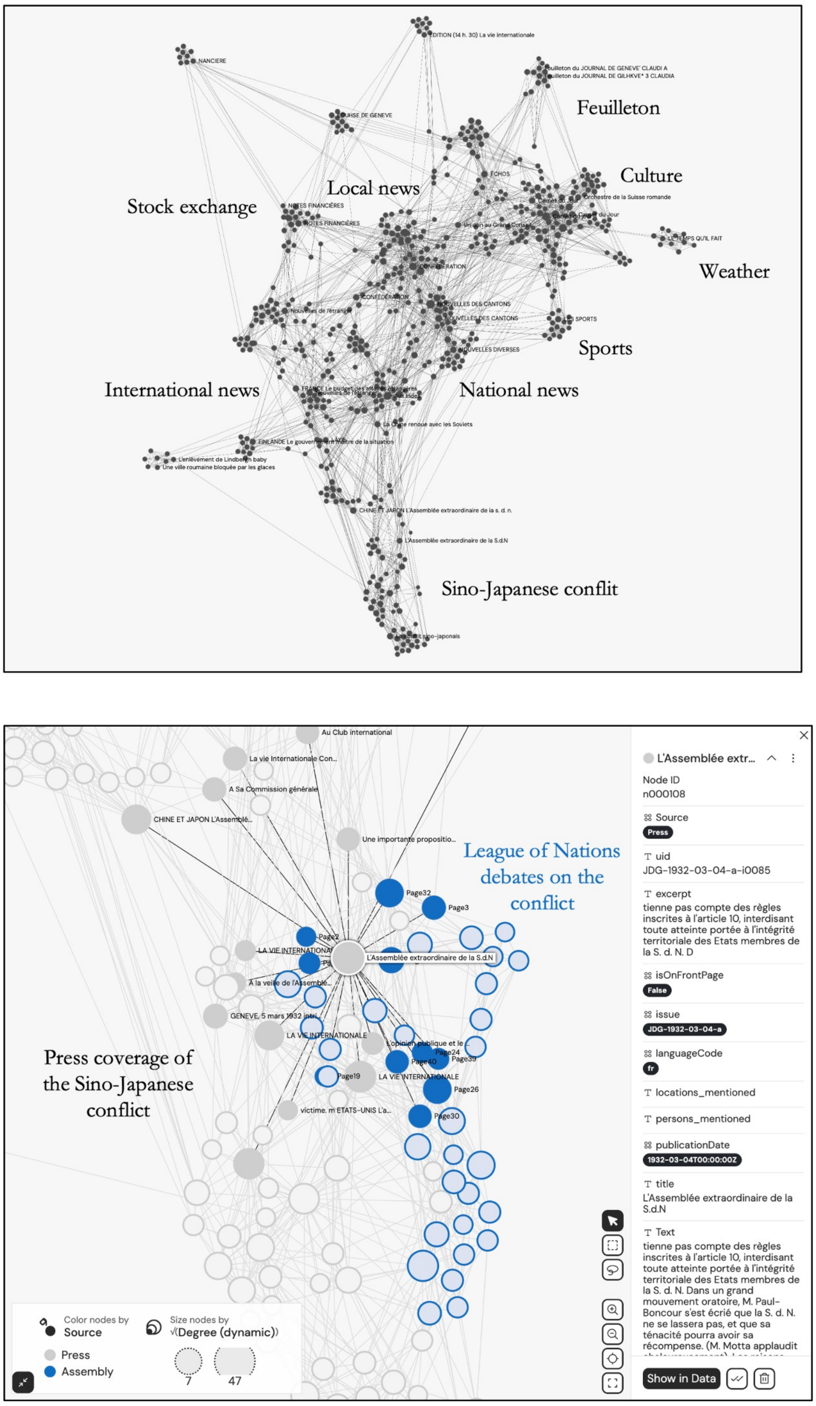

Figure 1. Top: Annotated network of (7) nearest neighbours for all articles in the *Journal de Genève* from March 1 to 10, 1932 (when the League of Nations debates the Sino-Japanese conflict). Bottom: Same network of press articles, but with the addition of the minutes of the League of Nations Assembly for the same days (in blue). Screenshots from the Gephi Lite interface (Girard et al., 2025).